Probability Theory, Decision Trees & Option Pricing

Thursday, November 5, 2020 7:35 PM

Historical Returns – Risk/Reward

PT – Opt
CAPM/FF3 – Reg] → Price Risk

Hedge Risk → Options

Probability Theory

1. $P(\text{Everything}) = 1$

2. $P(\text{Equally likely } A) = \dfrac{\#\ A \text{ occur}}{\#\ \text{Outcomes}}$

Flip Coin H,T $\quad P(H) = \dfrac{1}{2}$

Roll Die 1, 2, 3, 4, 5, 6 $\quad P(1) = \dfrac{1}{6}$

$P(\text{odd}) = \dfrac{\#\ \text{odd}}{\#\ \text{outcomes}} = \dfrac{3}{6}$

3. $P(A^c) = 1 - P(A)$

$P(\text{Not rolling } 1) = 1 - P(1)$
$= 1 - \frac{1}{6}$
$= \frac{5}{6}$

$\boxed{2, 3, 4, 5, 6} = \dfrac{5}{6}$

4. Conditional Probabilities

$P(A \text{ given } B)$

$P(1) = \dfrac{1}{6}$ $\quad$ 1 ~~2~~ 3 ~~4~~ 5 ~~6~~

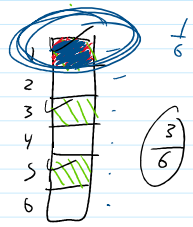
$P(1 \text{ given odd}) = \dfrac{1}{3}$

$P(1 \text{ given even}) = 0$ $\quad$ 1 $\quad$ 2, 4, 6

$$\boxed{P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}}$$

$P(1 \text{ given odd}) = \dfrac{P(1 \text{ and odd})}{P(\text{odd})}$

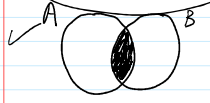
$= \dfrac{\frac{1}{6}}{\frac{3}{6}} = \dfrac{1}{3}$



$\frac{1}{6}$

$\frac{3}{6}$

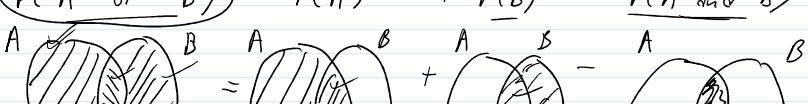
5. $P(A \text{ and } B) = P(A \text{ given } B) \times P(B)$

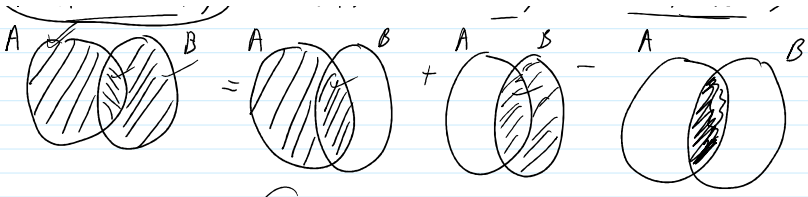


$P(1 \text{ and odd}) = P(1 \text{ given odd}) \times P(\text{odd})$

$\frac{1}{6} = \frac{1}{3} \times \frac{1}{2}$

6. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

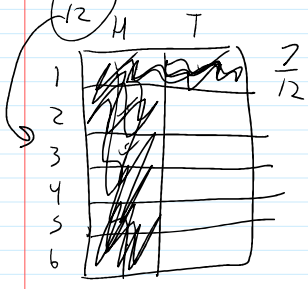




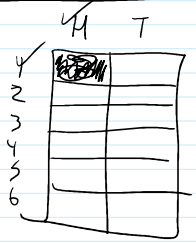
$P(1 \text{ or } H) = P(1) + P(H) - P(1 \text{ and } H)$

$\frac{7}{12}$

$\frac{1}{6} + \frac{1}{2} - \frac{1}{12} = \frac{7}{12}$



$\frac{7}{12}$



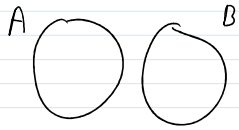
$\frac{1}{12}$

Independent $P(A \text{ given } B) = P(A)$

$P(1) = \frac{1}{6}$

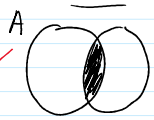
$P(1 \text{ given } H) \overset{\text{Ind}}{=} P(1) = \frac{1}{6}$

Mutually Exclusive $P(A \text{ and } B) = 0$



$P(H \text{ and } T) = 0$

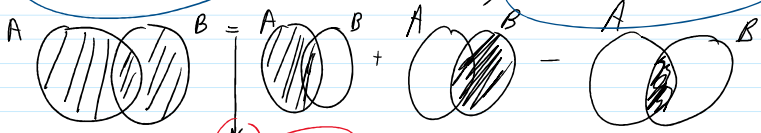
1. $P(A \text{ and } B) = P(A \text{ given } B) \times P(B)$



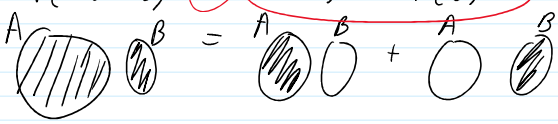
$\overset{\text{Ind}}{=} P(A) \times P(B)$

Ind $\Rightarrow P(A \text{ given } B) = P(A)$

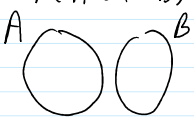
2. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$



$P(A \text{ or } B) \overset{\text{ME}}{=} P(A) + P(B)$

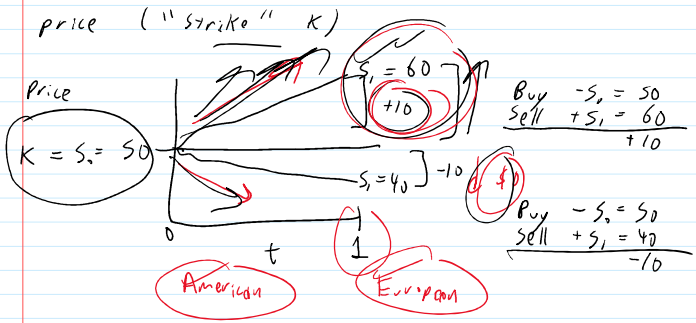


ME $\Rightarrow P(A \text{ and } B) = 0$



Option → Derivatives
↳ derives value from another asset

Call Option – gives the owner the right but not obligation to buy the stock @ predetermined price ("strike" K)



Buy $-S_0 = 50$
Sell $+S_1 = 60$
$+10$

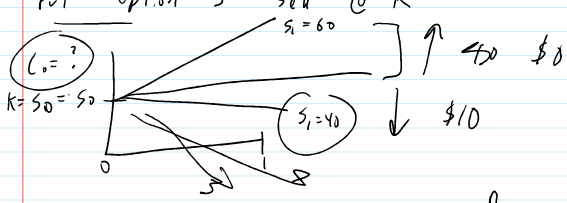
Buy $-S_0 = 50$
Sell $+S_1 = 40$
-10

up Execute option Buy $-K = 50$
Sell $+S_1 = 60$
$+10$

Down ~~Execute option~~ ~~Buy $-K = 50$ Sell $+S_1 = 40$ -10~~

Not execute option → \$0

Put Option → Sell @ K



↑ 40 \$0

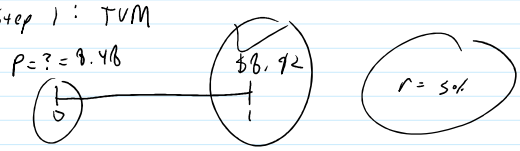
↓ \$10

up ~~Buy $-S_1 = 60$ Sell $+K = 50$ -10~~

P_0 Not execute \$0

Down Buy $-S_1 = 40$
Sell $+K = 50$
$+10$

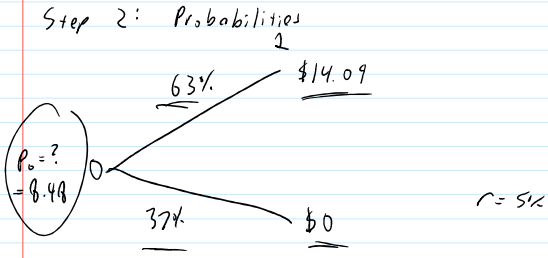
Step 1: TVM



$r = 5\%$

$$PV = \frac{FV}{(1+r)^t} = \frac{8.92}{(1+5\%)^1} \approx 8.48$$

Step 2: Probabilities



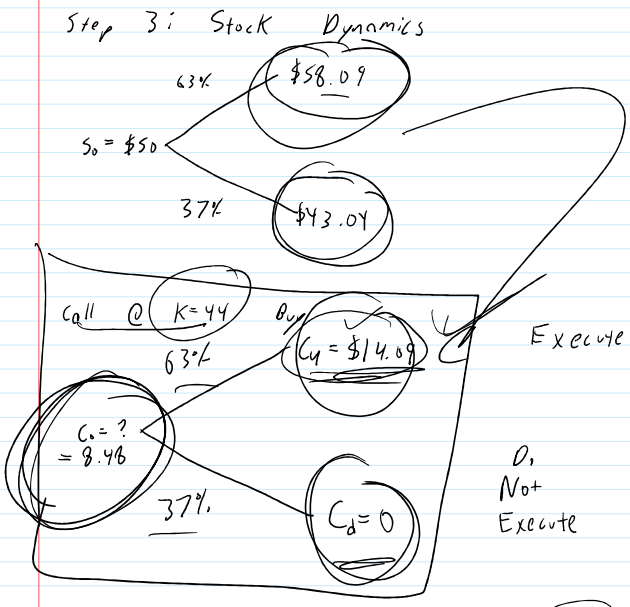
$r = 5\%$

$E[CF_1] = 63\% \times \$14.09 + 37\% \times \$0$

$= \$8.92$

$PV = \frac{E[CF_1]}{(1+r)^1} = \frac{8.92}{(1+5\%)^1} = 8.48$

Step 3: Stock Dynamics



Up $14 - 8 = 6$

Down $0 - 8 = -8$

Execute

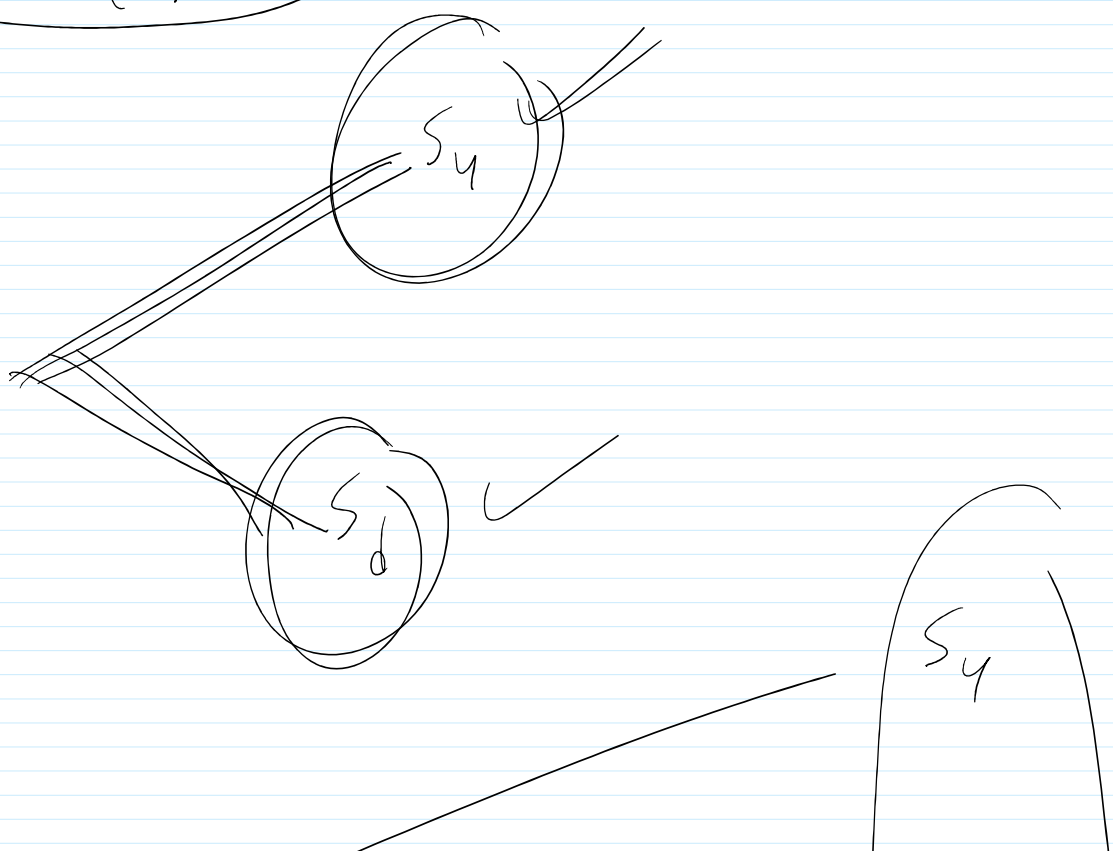
Buy @ K = 44
Sell @ $S_1 = 58$
$+\$14.09$

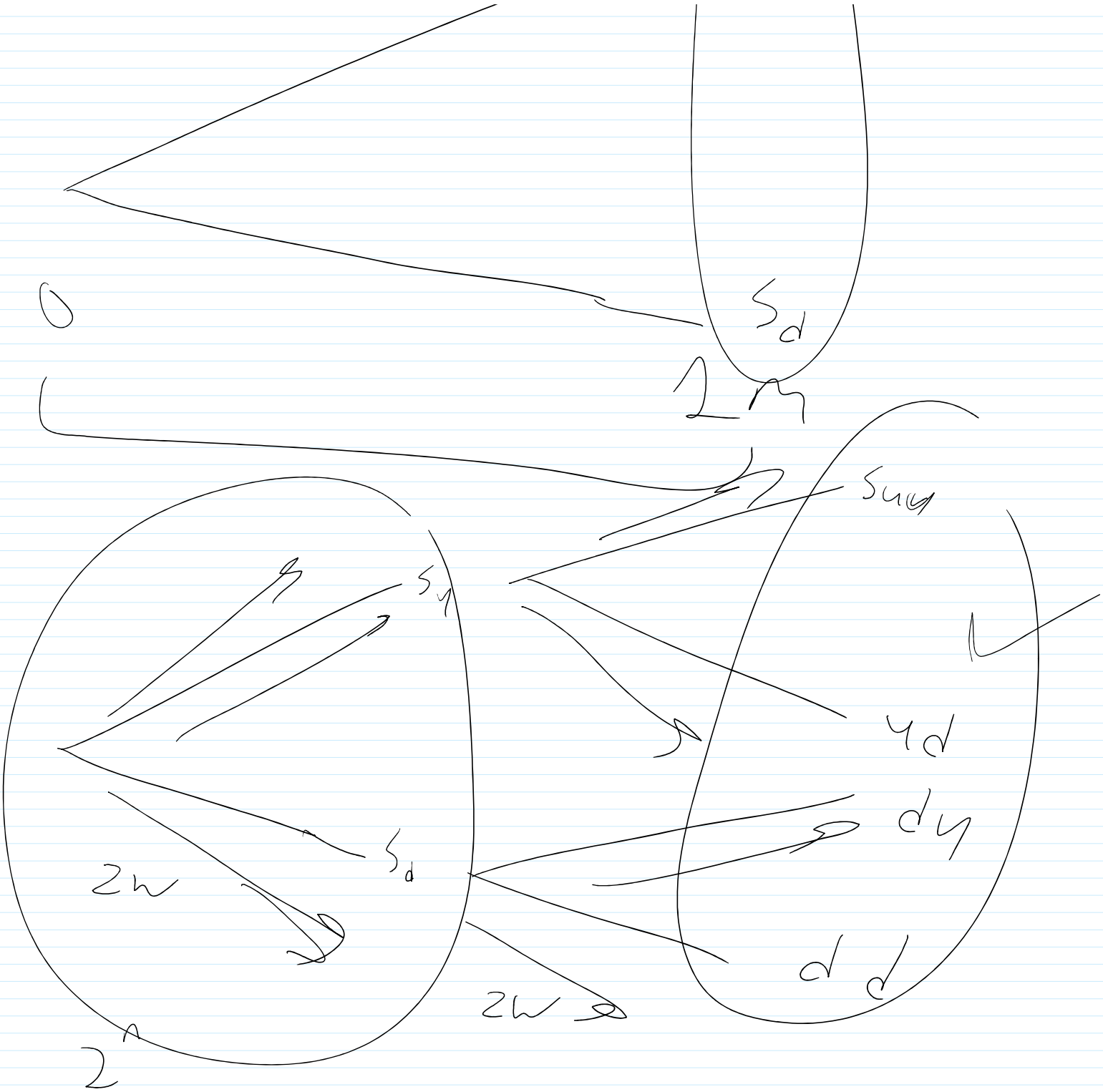
C_1 Not Execute

Buy K = 44
Sell $S_1 = 43$
-1

$E[C_1] = 63\% \times \$14.09 + 37\% \times \$0 = \$8.92$

$C_0 = \frac{E[C_1]}{(1+r)^1} = \frac{8.92}{(1+5\%)^1} = 8.49$




0
s_d
$2m$
s_{uw}
s_y
y_d
d_y
s_d
$2w$
d_d
$2w$
2^n